

MedRef v5.0 Multi-Authority Architecture

Dual-Region Clinical Evidence Engine (US FDA + CDSCO India)

Executive Summary & Strategic Vision

MedRef v5.0 Multi-Authority Architecture elevates MedRef from a US-focused drug prescribing search engine into a region-aware clinical evidence platform. By establishing dual regulatory channels—combining **US Sources** (DailyMed, openFDA, RxNorm, NIH, CDC) and **India Sources** (CDSCO, National Formulary of India, Indian Pharmacopoeia, ICMR, NTEP, NACO, MoHFW)—MedRef seamlessly bridges Western prescribing labels with Indian brand names, Fixed-Dose Combinations (FDCs), Schedule H/H1/X classifications, and regional clinical practice guidelines.

1. Multi-Authority Data Source Mapping

Region	Regulatory Authorities & Sources	Clinical Scope & Value
United States	• DailyMed (Authoritative SPL XML) • openFDA (Metadata & Recalls) • RxNorm (Normalized Clinical Drugs) • NIH & CDC (Clinical Research & Guidelines)	Full US prescribing labels, boxed warnings, molecular structures, standardized LOINC section mapping.
India	• CDSCO (Central Drugs Standard Control) • NFI (National Formulary of India) • IP (Indian Pharmacopoeia) • ICMR / NTEP / NACO / MoHFW (National Guidelines) • PvPI (Pharmacovigilance Programme)	Indian brand resolution (Dolo 650, Novamox, Telma), Fixed-Dose Combinations (FDCs), Schedule H/H1/X regulatory status, regional treatment guidelines.

2. 15 Strategic Real-World Use Cases for CDSCO & India Data

#	Use Case Feature	Clinical Problem Solved & Operational Flow
1	Indian Brand Resolution	Maps Indian trade names (<i>Novamox, Augmentin, Dolo 650, Taxim, Monocef, Pan-D, Zifi, Telma, Azee</i>) → Generic profile → DailyMed/CDSCO label.
2	India-Only Drug Approvals	Handles drugs approved by CDSCO but not FDA (e.g. <i>Saroglitazar</i>), preventing false 'unknown drug' errors.
3	Fixed-Dose Combinations (FDCs)	Indexes multi-component Indian FDCs (<i>Telmisartan + Amlodipine, Amoxicillin + Clavulanate, Metformin + Vildagliptin</i>).
4	CDSCO Safety Alerts & Recalls	Surfaces real-time CDSCO batch recalls, alert notices, and label revisions directly on the drug profile.
5	Multi-Region Approval Status	Displays side-by-side approval status across FDA (US), CDSCO (India), and EMA (Europe).
6	Schedule Classification	Classifies Indian prescription schedules: Schedule H, Schedule H1 (restricted antibiotics), and Schedule X (narcotics).
7	Multi-Region Pregnancy Guidance	Compares US FDA pregnancy/lactation warnings with CDSCO & ICMR maternal safety guidelines.
8	Indian Manufacturer Insights	Surfaces generic manufacturers (<i>Sun Pharma, Cipla, Dr Reddy's, Torrent, Mankind, Lupin</i>) for clinical availability.
9	Generic Equivalent Finder	Helps clinicians locate available bioequivalent generic brands in India.
10	Drug Shortage Tracking	Monitors US FDA Drug Shortages alongside CDSCO / Indian domestic supply chain shortages.
11	Indian Clinical Guidelines	Indexes ICMR, NTEP (Tuberculosis), NACO (HIV), RSSDI (Diabetes), and CSI/API (Cardiology) guidelines.
12	Indian Dosing & Regimens	Surfaces national treatment regimens for Tuberculosis (NTEP), Malaria, Kala-Azar, and Snake Bite management.
13	Adverse Event Reporting (PvPI)	Integrates Pharmacovigilance Programme of India (PvPI) reporting alongside FDA FAERS.
14	Indian OTC & Restricted Status	Differentiates Over-The-Counter vs Prescription vs Hospital-Only restricted drugs under Indian law.
15	Regional Conversational Chat	End-to-end multi-authority chat flow: Brand → Generic → CDSCO → DailyMed → ICMR → Grounded Answer.

3. Multi-Collection Vector Architecture & Regional Retrieval Flow

MedRef v5.0 replaces single collection storage with 10 specialized Qdrant vector collections routed by an Intelligent Intent Router and Country Context Manager:

User Query (e.g., 'Novamox dose for sinusitis' or 'Telmisartan + Amlodipine FDC') ■ ▼ Intent Router (Classifies: Brand Lookup, FDC Search, Guideline Query, Safety Check) ■ ▼ Country Context Manager ■■■■ US Context → Route to [drug_labels_us] (DailyMed, FDA, RxNorm) ■■■■ India Context → Route to [drug_labels_india] + [disease_guidelines] (CDSCO, NFI, ICMR, NTEP) ■■■■ Global Context → Cross-retrieve US & India collections side-by-side ■ ▼ Multi-Collection Vector Retrieval & Cross-Encoder Reranking ■ ▼ Grounded Response Generation + Multi-Authority Trust Badges [FDA]

[DailyMed] [CDSCO] [ICMR] [NTEP] [ADA] [KDIGO]

4. Target Collection Structure

Qdrant Collection	Indexed Content & Domain	Metadata Schema Keys
drug_labels_us	500 US Prescribing Labels (DailyMed/openFDA)	drug_name, section, authority='DailyMed'
drug_labels_india	CDSCO Approvals, NFI & IP Formularies	brand_name, generic_name, schedule, authority='CDSCO'
drug_interactions	Pairwise Drug-Drug & FDC Interactions	drug_a, drug_b, severity, mechanism
disease_guidelines	US (ADA, GOLD, GINA) & India (ICMR, NTEP, NACO)	guideline_body, disease, recommendation_grade
drug_regulatory	Schedules H/H1/X, Recalls, Safety Alerts	alert_type, release_date, affected_batches
brand_aliases	US & Indian Trade Name Mapping (1,500+ aliases)	brand_name, generic_name, country_code